Iglesia lecture

Tuesday, 3 October 2023 10:29 AM

$C \equiv O + O=O \rightarrow O=C=O$

+ Breaking into smaller steps (elementary steps) w/ smaller

+ Dimensions = vibrational modes of a molecule $(3N-1)$ } Becomes an N-dimensional space problem

Mechanism-Based Chemical Kinetics

i) Determine kinetic response "reactive condition" (P_j, T)
↳ transport artefact
(probe away from actual active site so need to account for it)

$f(P_j, T)$

↳ Thermodynamics

$A \rightarrow B$ ← measure the rate of only forward reaction (isolate primary & secondary reactions)

2) Propose : elementary steps
↳ Identify reactive (& less reactive) species
↳ PSSH
(pseudo ss) } eliminate the need to know what you cannot see
↳ QE
↳ MASI (limited landing place) → empty, reactant, product

3) Assess reasonable : (seeking contradictions)
↳ Consistent w/ thermodynamics, calculations
(TST) → Thermo construct, theoretical in nature

↳ Direct observations (spectroscopic method)

$CH_4 + H_2O \rightarrow 3H_2 + CO$

Langmuir Hinshelwood reaction
vs.
Eyling ...

Elementary step } "Most simple"

vs.

Elementary step (when do you stop breaking it down) } "Most simple"

$aA + bB \rightleftharpoons cC + dD$

Eyling ...

↓

Has to occur as written! (can extract rate information directly) } Can use law of mass action

$$\frac{k_1}{k_{-1}} = K_1$$

$(*) + (A*) \rightleftharpoons (B*) + (C*)$

$\vec{r}_1 = k_1 (A*)(*) = k_1 (A*)\,\theta_*$

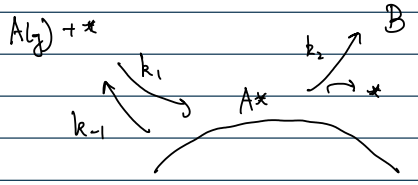
(L) → total no. of squares

} Langmuirian treatment (catalyst is checkerboard)
+ Identical binding
+ Species are mobile → Langmuir (worked for GE)

↓

an approximation

$A \rightarrow B$



σ_i → No. of times that step needs to take place

↑ stoichiometric constant

$A + * \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A*$

$A* \xrightarrow{k_2} B + *$

$(L) = (*) + (A*)$

$$\frac{d(A)}{dt} = -k_1 (A)(*) + k_{-1} (A*)$$

$$\frac{d(A*)}{dt} = k_1 (A)(*) - (k_{-1} + k_2)(A*)$$

using PSSH this term cannot change w/ time $\simeq 0$